

List of Unix Commands Used

1. Create Working Directory

This command creates a folder where all experiment files will be stored.

```
mkdir purchase_analysis
```

2. Navigate to the Working Directory

This command moves into the created directory.

```
cd purchase_analysis
```

3. Check Current Directory

Displays the current working directory.

```
pwd
```

4. Create Dataset File

Creates and opens the dataset file for editing.

```
nano purchases.txt
```

After executing this command, the purchase records are entered manually and saved.

5. Display Dataset Content

Displays the entire dataset file.

```
cat purchases.txt
```

Task 2: Basic Dataset Analysis

6. Count Total Purchase Records

Counts the total number of lines (purchase records) in the dataset.

```
wc -l purchases.txt
```

7. Display First Few Records

Displays the first 10 records of the dataset.

```
head purchases.txt
```

8. Display Only Product Column

Extracts the **Product Name column (2nd column)** from the dataset.

```
cut -d " " -f2 purchases.txt
```

Explanation:

- `cut` → extracts columns
 - `-d " "` → space delimiter
 - `-f2` → second column
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Task 3: Category-wise Purchase Analysis

9. Extract Category Column

Extracts the **Category column (3rd column)**.

```
cut -d " " -f3 purchases.txt
```

10. Sort Category Values

Sorts the category values alphabetically.

```
cut -d " " -f3 purchases.txt | sort
```

Explanation:

- `|` → pipe operator passes output to next command
 - `sort` → arranges data alphabetically
-

11. Count Purchases per Category

Counts the number of occurrences of each category.

```
cut -d " " -f3 purchases.txt | sort | uniq -c
```

Explanation:

- `uniq -c` → counts repeated values
-

Task 4: Most Frequently Purchased Products

12. Extract Product Column

Extracts all product names.

```
cut -d " " -f2 purchases.txt
```

13. Count Product Frequency

Counts how many times each product appears.

```
cut -d " " -f2 purchases.txt | sort | uniq -c
```

14. Sort Products by Frequency (Descending Order)

Sorts the results from highest to lowest frequency.

```
cut -d " " -f2 purchases.txt | sort | uniq -c | sort -nr
```

Explanation:

- `sort -nr`
 - `-n` → numeric sorting
 - `-r` → reverse order (largest first)
-

15. Display Top 3 Most Purchased Products

Shows the top three products with the highest purchase frequency.

```
cut -d " " -f2 purchases.txt | sort | uniq -c | sort -nr | head -3
```

Explanation:

- `head -3` → displays first 3 results
-

Complete Command Pipeline Summary

The **most advanced pipeline used in this activity** is:

```
cut -d " " -f2 purchases.txt | sort | uniq -c | sort -nr | head -3
```

This pipeline performs the following operations:

1. Extracts product names
2. Sorts them
3. Counts occurrences
4. Sorts by frequency
5. Displays the top 3 products

Unix Commands Used in the Experiment

Command	Purpose
mkdir	Create directory
cd	Change directory
pwd	Display current directory
nano	Create/edit file
cat	Display file content
wc	Count lines/words
head	Display first lines
cut	Extract columns
sort	Sort text
uniq	Remove duplicates and count
head -3	Display top results
pipe ()	Combine multiple commands

Total Commands Used

12 Unix commands

mkdir
cd
pwd
nano
cat
wc
head
cut
sort
uniq
head -3